

it has been shown to reduce alcohol consumption on average by 60%

- Reduction in alcohol consumption has been shown to be maintained for 12 years
- There is no clinically reported dose-dependent hepatotoxicity
- Like naltrexone, nalmefene is a mu opioid receptor antagonist; however, nalmefene is also a partial agonist at kappa opioid receptors, which are thought to contribute to the dysphoria and anxiety experienced during alcohol withdrawal
- Nalmefene is unique in that it is taken as needed when the patient perceives a risk of drinking alcohol

Suggested Reading

Gual A , He Y , Torup L , et al. A randomized, double-blind, placebo-controlled, efficacy study of nalmefene, as-needed use, in patients with alcohol dependence . *Eur Neuropsychopharmacol* 2013 ;**23** :1432 –42.

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Mann K , Bladstrom A , Torup L , Gual A , van den Brink W. Extending the treatment options in alcohol dependence: a randomized controlled study of as-needed nalmefene . *Biol Psychiatry* 2013 ;**73** (8):706–13 .

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Rosner S , Hackl-Herrwerth A , Leucht S , et al. Opioid antagonists for alcohol dependence . *Cochrane Database Syst Rev* 2010 Dec;**8** (12):CD001867 .

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van den Brink W , Aubin HJ , Bladstrom A , et al. Efficacy of as-needed nalmefene in alcohol-dependent patients with at least a high drinking risk level: results from a subgroup analysis of two randomized controlled 6-month studies . *Alcohol Alcohol* 2013 ;**48** (5):570–8 .

Naltrexone

Therapeutics

Brands

- Revia (oral)
- Vivitrol (injection)
- *see index for additional brand names*

Generic?

Yes (not injection or solution)

Class

- Neuroscience-based Nomenclature: opioid receptor antagonist (O-RAn)
- Alcohol dependence treatment; mu opioid receptor antagonist

Commonly Prescribed for

(bold for FDA approved)

- **Alcohol dependence**
- **Blockade of effects of exogenously administered opioids (oral)**
- **Prevention of relapse to opioid dependence (injection)**

How the Drug Works

- Blocks mu opioid receptors, preventing exogenous opioids from binding there and thus preventing the pleasurable effects of opioid consumption
- Reduces alcohol consumption through modulation of opioid systems, thereby reducing the reinforcing effects of alcohol

How Long Until It Works

- Can begin working within a few days but maximum effects may not be seen for a few weeks

If It Works

- Reduces alcohol and opioid consumption by diminishing their reinforcing properties (rewarding effects, cravings)

If It Doesn't Work

- Evaluate for and address contributing factors
- Consider switching to another agent
- Consider augmenting with acamprosate

Best Augmenting Combos for Partial Response or Treatment Resistance

- Acamprosate
- Augmentation therapy may be more effective than monotherapy

- Augmentation with behavioral, education, and/or supportive therapy in groups or as an individual is probably key to successful treatment

Tests

- Urine screen for opioids and/or naloxone challenge test prior to initiating treatment for opioid use
- None for use in treating alcohol dependence, although baseline liver function testing, usually obtained anyway for managing alcohol dependence, may be useful

Side Effects

How Drug Causes Side Effects

- Blockade of mu opioid receptors

Notable Side Effects

- Nausea, vomiting, decreased appetite
- Dizziness, dysphoria, anxiety
- Injection site reactions (pain, tenderness, pruritis, induration, swelling, erythema, or bruising); in some cases injection site reactions may be very severe



Life-Threatening or Dangerous Side Effects

- Eosinophilic pneumonia

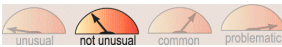
- Hepatocellular injury (at excessive doses)
- Severe injection site reactions requiring surgery

Weight Gain



- Reported but not expected

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Adjust dose
- If side effects persist, discontinue use

Best Augmenting Agents for Side Effects

- Dose reduction or switching to another agent may be more effective since most side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Oral: 50 mg/day or 100 mg on Mon and Wed and 150 mg on Fri
- Injection: 380 mg every 4 weeks

Dosage Forms

- Tablet 25 mg, 50 mg, 100 mg
- Methylbuprenorphine hydrochloride tablet 150 mg
- Solution (subcutaneous) 8 mg/0.4 mL, 12 mg/0.6 mL
- Intramuscular formulation 380 mg/vial

How to Dose

- Patient should be opioid-free for 7–10 days prior to initiating treatment with buprenorphine, as confirmed by negative urine screen and/or naloxone challenge test
- For treating alcohol dependence (oral): recommended dose is 50 mg/day; titration is not required
- For treating alcohol dependence (injection): 380 mg delivered intramuscularly in the gluteal region every 4 weeks; alternate buttocks; must be administered by health-care professional
- For treating opioid dependence (oral): initial 25 mg/day; day 2 can increase to 50 mg/day
- For treating opioid dependence (injection): 380 mg delivered intramuscularly in the gluteal region every 4 weeks or once a month; alternate buttocks; must be administered by healthcare professional

Dosing Tips

- Providing educational materials and counseling in combination with naltrexone treatment can increase the chances of success
- Individuals who abstain from alcohol for several days prior to initiating treatment with naltrexone may have greater reductions in the number of drinking days as well as number of heavy drinking days, and may also be more likely to abstain completely throughout treatment
- Long-acting naltrexone (injection) must be kept refrigerated
- For long-acting naltrexone (injection), the suspension should be administered immediately after mixing
- Adherence greatly enhanced and assured for 30 days at a time when administering by once monthly depot injection; oral treatment requires 30 decisions to comply in 30 days whereas long-acting injection requires only one decision every 30 days to comply
- Patients must be completely withdrawn and abstinent from opioids for 5 days (short-acting opioids) to 7 days (long-acting opioids) before starting naltrexone

Overdose

- Nausea, abdominal pain, sedation, dizziness, injection site reactions

Long-Term Use

- Has been studied in trials up to 1 year

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Elimination half-life of oral naltrexone is approximately 13 hours
- Elimination half-life of naltrexone via injection is 5–10 days



Drug Interactions

- Is hepatically metabolized by dihydrodiol dehydrogenase and not by the CYP450 enzyme system, and thus is unlikely to be affected by drugs that induce or inhibit CYP450 enzymes
- Can block the effects of opioid-containing medications (e.g., some cough and cold remedies, antidiarrheal preparations, opioid analgesics)
- Concomitant administration with acamprosate may increase plasma levels of acamprosate, but this does not appear to be clinically significant and dose adjustment is not recommended



Other Warnings/Precautions

- Can cause hepatocellular injury when given in excessive doses

- To prevent withdrawal in patients dependent on opioids, patients must be opioid-free for at least 7–10 days prior to initiating treatment
- Attempts by patients to overcome blockade of opioid receptors by taking large amounts of exogenous opioids could lead to opioid intoxication or even fatal overdose
- Individuals who have been previously treated with naltrexone should be warned that they may respond to lower doses of opioids than previously used, and thus previous doses may also lead to opioid intoxication
- Individuals receiving naltrexone who require pain management with opioid analgesia may need a higher dose than usual and may experience deeper and more prolonged respiratory depression; pain management with non-opioid or rapid-acting opioid analgesics is recommended if possible
- Monitor patients for emergence of depressed mood or suicidality
- Use cautiously in individuals with known psychiatric illness
- Injection should be used cautiously in individuals with thrombocytopenia or any coagulation disorder

Do Not Use

- If patient is taking opioid analgesics
- If patient is currently dependent on opioids or is in acute opiate withdrawal

- If patient has failed the naloxone challenge test or has a positive urine screen for opioids
- If patient has acute hepatitis or liver failure
- If there is a proven allergy to naltrexone
- If there is a proven allergy to polylactideco-glycolide (PLG), arboxymethylcellulose, or any other components of the diluent (injection)

Special Populations

Renal Impairment

- Dose adjustment not generally necessary for mild impairment
- Not studied in moderate to severe renal impairment

Hepatic Impairment

- Has the potential to cause hepatocellular injury when given in excessive doses
- Dose adjustment not generally necessary for mild impairment
- Not studied in severe hepatic impairment
- Contraindicated in acute hepatitis or liver failure

Cardiac Impairment

- Limited available data

Elderly

- Safety and efficacy have not been established
- Some patients may tolerate lower doses better



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Pregnant women needing to stop drinking may consider behavioral therapy before pharmacotherapy
- Not generally recommended for use during pregnancy, especially during first trimester

Breast Feeding

- Some drug is found in mother's breast milk
- Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Individuals who are not ready to abstain completely from alcohol
- For binge drinkers

Potential Disadvantages

- Patients who “drink over” their treatment, including their long-acting injection
- Less effective in patients who are not abstinent at the time of treatment initiation

Primary Target Symptoms

- Alcohol dependence

Pearls

- Not only increases total abstinence, but also can reduce days of heavy drinking
- May be a preferred treatment if the goal is reduced-risk drinking (i.e., 3–4 drinks per day in men, maximum 16 drinks per week; 2–3 drinks per day in women, maximum 12 drinks per week)
- Less effective in patients who are not abstinent at the time of treatment initiation

- Some patients complain of apathy or loss of pleasure with chronic treatment
- The combination of naltrexone and bupropion has demonstrated efficacy as treatment for obesity and is currently being evaluated in a long-term study to assess the cardiovascular health outcomes of this treatment

Suggested Reading

Anton RF , O'Malley SS , Ciraulo DA , et al. Combined pharmacotherapies and behavioral interventions for alcohol dependence: the COMBINE study: a randomized controlled trial . *JAMA* 2006 ;**295** (17):2003 –17.

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Johansson BA , Berglund M , Lindgren A . Efficacy of maintenance treatment with naltrexone for opioid dependence: a meta-analytical review . *Addiction* 2006 ;**101** (4):491 –503 .

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Mannelli P , Peindl K , Masand PS , Patkar SS . Long-acting injectable naltrexone for the treatment of alcohol dependence . *Expert Rev Neurother* 2007 ;**7** (10):1265 –77.

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Rosner S , Leucht S , Leher P , Soyka M . Acamprosate supports abstinence, naltrexone prevents excessive drinking: evidence from a meta-analysis with unreported outcomes . *J Psychopharmacol* 2008 ;**22** :11 –23 .

Naltrexone/Bupropion

Therapeutics

Brands

- Contrave
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: opioid receptor antagonist (naltrexone, O-RAn) and dopamine reuptake inhibitor and releaser (bupropion, D-RIRe)
- Opioid antagonist (naltrexone) combined with a norepinephrine dopamine reuptake inhibitor (bupropion); weight management medication

Commonly Prescribed for

(bold for FDA approved)

- **Chronic weight management (adjunct to reduced-calorie diet and increased physical activity) in adults with an initial BMI of**

at least 30 kg/m² (obese) or at least 27 kg/m² (overweight) in the presence of at least 1 weight-related comorbid condition

How the Drug Works

- Bupropion increases dopamine and norepinephrine by blocking both the dopamine and the norepinephrine transporters. In the hypothalamus, these 2 neurotransmitters activate POMC neurons, causing the release of proopiomelanocortin (POMC). POMC is then broken down into alpha-melanocyte stimulating hormone, which binds to melanocortin 4 receptors to suppress appetite. However, stimulation of POMC neurons also activates an endogenous opioid-mediated negative feedback loop, which mitigates the appetite-suppressing effects.
- Naltrexone can block mu opioid receptors, thus preventing opioid-mediated negative feedback. Such an action would synergize with simultaneous activation of the appetite-suppressing pathway by bupropion. This results in more robust and long-lasting appetite suppression than with either drug alone

How Long Until It Works

- At least 5% weight loss is generally achieved after 12 weeks on maximum daily dose

If It Works

- Patients may achieve 5–10% reduction from baseline in body weight

If It Doesn't Work

- Discontinue if 5% weight loss is not achieved after 12 weeks on maximum daily dose



Best Augmenting Combos for Partial Response or Treatment Resistance

- Naltrexone/bupropion itself should be administered in conjunction with reduced-calorie diet and increased physical activity
- Often best to try another strategy prior to resorting to augmentation

Tests

- Measurement of blood glucose levels before and during treatment is recommended for patients with type 2 diabetes
- Measurement of blood pressure and heart rate prior to starting naltrexone/bupropion and during treatment is recommended, particularly in patients with cardiac or cerebrovascular disease

Side Effects

How Drug Causes Side Effects

- CNS side effects theoretically due to excessive actions at mu opioid receptors or excessive actions on norepinephrine and dopamine

Notable Side Effects

- Nausea, constipation, vomiting, diarrhea, dry mouth
- Headache, dizziness, insomnia



Life-Threatening or Dangerous Side Effects

- Increased blood pressure or heart rate
- Rare seizures (risk increases with doses above the recommended maximums; risk increases for patients with predisposing factors)
- Hepatocellular injury (at excessive doses)
- Rare induction of mania
- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)



Weight Gain

- Reported but not expected



Sedation

- Reported but not expected

What to Do About Side Effects

- Wait
- In a few weeks, switch to another agent

Best Augmenting Agents for Side Effects

- Often best to try another treatment prior to resorting to augmentation strategies to treat side effects
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require discontinuation of bupropion

Dosing and Use

Usual Dosage Range

- 16 mg naltrexone/180 mg bupropion twice per day

Dosage Forms

- Extended-release tablet (naltrexone/bupropion) 8 mg/90 mg

How to Dose

- Week 1: 8 mg/90 mg in the morning; week 2: increase to 8 mg/90 mg twice per day; week 3: increase to 16 mg/180 mg in the morning and 8 mg/90 mg in the evening; week 4: increase to 16 mg/180 mg twice per day (maximum recommended dose)
- Discontinue if at least 5% weight loss is not achieved after 12 weeks on 16 mg/180 mg twice per day

Dosing Tips

- Do not break or chew tablets as this will alter controlled-release properties
- Can be taken with food, but should not be taken with a high-fat meal because this may increase bupropion and naltrexone systemic exposure
- Patients must be completely withdrawn and abstinent from opioids for 5 days (short-acting opioids) to 7 days (long-acting opioids) before starting naltrexone-bupropion; if patients require intermittent opioid treatment, naltrexone/bupropion should be temporarily discontinued and lower doses of opioids may be needed
- If intolerable anxiety, insomnia, agitation, akathisia, or activation occur upon either dosing initiation or discontinuation, consider the possibility of activated bipolar disorder and switch to a mood stabilizer or an atypical antipsychotic

Overdose

- Naltrexone: nausea, abdominal pain, sedation, dizziness
- Bupropion: rarely lethal; seizures, cardiac disturbances, hallucinations, loss of consciousness

Long-Term Use

- Has been evaluated in controlled studies up to 56 weeks

Habit Forming

- No

How to Stop

- Tapering is prudent

Pharmacokinetics

- Mean elimination half-life following a single dose of naltrexone/bupropion to healthy subjects was approximately 5 hours for naltrexone and 21 hours for bupropion
- Bupropion inhibits CYP450 2D6



Drug Interactions

- Do not use with MAOIs, including 14 days after MAOIs are stopped
- Via CYP450 2D6 inhibition, naltrexone/bupropion could increase concentrations of drugs metabolized by CYP450 2D6 (e.g., TCAs, thioridazine); this may require lower doses of the concomitant medication
- Via CYP450 2D6 inhibition, bupropion could theoretically interfere with the analgesic actions of codeine, and increase the plasma levels of some beta blockers and of atomoxetine
- Bupropion is metabolized by CYP450 2B6; the maximum dose of naltrexone-bupropion in patients receiving a concomitant CYP450 2B6 inhibitor (ticlopidine, clopidogrel) is 8 mg/90 mg twice per day;

coadministration of naltrexone/bupropion with CYP450 2B6 inducers (e.g., ritonavir, lopinavir, efavirenz) is not recommended

- Bupropion should be used with caution in patients taking levodopa or amantadine, as these agents can potentially enhance dopamine neurotransmission and be activating
- Can block the effects of opioid-containing medications (e.g., some cough and cold remedies, antidiarrheal preparations, opioid analgesics)
- False-positive urine immunoassay screening tests for amphetamines have been reported in patients taking bupropion due to a lack of specificity of the screening tests
- False-positive results may be expected even following discontinuation of bupropion
- Naltrexone is hepatically metabolized by dihydrodiol dehydrogenase and not by the CYP450 enzyme system, and thus is unlikely to be affected by drugs that induce or inhibit CYP450 enzymes

Other Warnings/Precautions

- Naltrexone/bupropion can increase blood pressure or heart rate, particularly during the first 3 months of treatment
- Use cautiously with other drugs that increase seizure risk (TCAs, lithium, phenothiazines, thioxanthenes, some antipsychotics) or in

patients with predisposing factors that may increase the risk of seizures (e.g., history of head trauma)

- To prevent withdrawal in patients dependent on opioids, patients must be opioid-free (including tramadol) for at least 7–10 days prior to initiating treatment
- Attempts by patients to overcome blockade of opioid receptors by taking large amounts of exogenous opioids could lead to opioid intoxication or even fatal overdose
- Individuals who have previously been treated with naltrexone should be warned that they may respond to lower doses of opioids than previously used, and thus previous doses may also lead to opioid intoxication
- Weight loss may increase the risk of hypoglycemia in patients with type 2 diabetes mellitus treated with insulin and/or insulin secretagogues, and medication adjustments may be necessary
- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent
- Warn patients about the possibility of activation of suicidal ideation and advise them to report such side effects immediately

Do Not Use

- If patient has uncontrolled hypertension
- If patient is pregnant
- If patient has history of seizures

- If patient is anorexic or bulimic, either currently or in the past
- If patient has acute hepatitis or liver failure
- If patient is abruptly discontinuing alcohol or sedative use
- If patient is taking opioid analgesics, is currently dependent on opioids, or is in acute opiate withdrawal
- If patient is taking an MAOI
- If patient is taking Wellbutrin, Zyban, Aplenzin, or any other formulation that contains bupropion
- If patient is taking Revia, Vivitrol, or any other formulation that contains naltrexone
- If there is a proven allergy to naltrexone or bupropion

Special Populations

Renal Impairment

- Maximum dose is 8 mg/90 mg twice per day in patients with moderate to severe impairment
- Not studied or recommended for use in patients with end-stage renal disease

Hepatic Impairment

- Maximum dose is 8 mg/90 mg once per day in the morning

Cardiac Impairment

- Not systematically evaluated in patients with cardiac impairment
- Measure blood pressure and heart rate before and during treatment

Elderly

- Some patients may tolerate lower doses better



Children and Adolescents

- Safety and efficacy have not been established
- Not recommended for use in children or adolescents



Pregnancy

- Contraindicated
- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

Breast Feeding

- Some drug is found in mother's breast milk
- Recommended to discontinue drug or bottle feed